3. Real functions - Continuity

Calculus

• **Definition (Accumulation point and isolated point).** Let $D \subseteq \mathbb{R}$ and $p \in \mathbb{R} \cup \{-\infty, +\infty\}$.The point p is called an *accumulation point* of D if there exists $(x_n) \subseteq D \setminus \{p\}$ such that $x_n \to p$ if $n \to \infty$. A point p is an *isolated point* of D if $p \in D$ and p is not an accumulation point of D.

• **Definition (Continuity).**[1] Let $D \subseteq \mathbb{R}$, $p \in D$, and $f : D \to \mathbb{R}$. We say that f is continuous at the point $p$if either p is an isolated point of D or p is an accumulation point of D and, for all $\varepsilon > 0$, there exists $r > 0$ such that

$$|f(x) - f(p)| < \varepsilon \qquad \text{if} \qquad |x - p| < r.$$

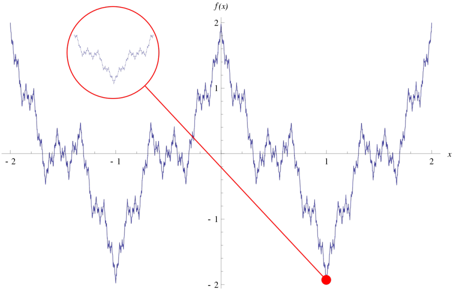
• Remark and definition: If a function is not continuous at a point in its domain, that point is called a *place of discontinuity*.

• **Definition (Global continuity).** Let $D \subseteq \mathbb{R}$ and $f : D \to \mathbb{R}$. We say that f is continuous on D if it is continuous at any point of D.

[1] Weierstrass (1815-1897) and Jordan (1838-1922).

Continuous functions

• *Continuity is often illustrated in the following way:* Continuous functions are those whose graphs could be drawn without lifting the pencil. This is not a lucky approach (see Weierstrass's function):



• *A better approach is:* If a function is continuous at an accumulation point of its domain, then moving "a little" from that point will change the value of the function only "a little".

• **Example (The parabola).** The function $f : \mathbb{R} \to \mathbb{R}$, $f(x) := x^2$ is continuous on its domain.

Proof. Let $p \in \mathbb{R}$ be any point. This is the point at which we prove continuity. Let $\varepsilon > 0$. Then

$$\begin{aligned} |f(x) - f(p)| &= |x^2 - p^2| = |(x - p + p)^2 - p^2| = |(x-p)^2 + 2p(x-p) + p^2 - p^2| \\ &= |(x-p)^2 + 2p(x-p)| \leq |(x-p)^2| + |2p(x-p)| \\ &= |x-p|^2 + 2|p| \cdot |x-p| =: (\star) \end{aligned}$$

In the sequel we assume that $|x - p| < 1$:

$$(\star) < |x-p| + 2|p| \cdot |x-p| = (1 + 2|p|) \cdot |x-p|.$$

Thus let r be any element of the interval

$$\Big]0, \min\Big(\frac{\varepsilon}{1+2|p|}, 1\Big)\Big[.$$

Continuity and convergent sequences

• **Theorem.** A function $f : D \to \mathbb{R}$ is continuous at an accumulation point $p \in D$ if and only if, for any real sequence $(x_n) \subseteq D$ converging to p, the real sequence $(f(x_n)) \subseteq \mathbb{R}$ is convergent and its limit is $f(p)$.

• Note that the theorem above states precisely that continuous functions are interchangeable with limit:

$$\lim_{n\to\infty} f(x_n) = f(p) = f\big(\lim_{n\to\infty} x_n\big).$$

• Motivated by the above theorem, continuity can be illustrated as follows: *Continuous image of a convergent real sequence is convergent.*

• Using the theorem above, it is usually easy to show that a function is not continuous at a given point.

• **Fact.** The set of rational numbers and the set of irrational numbers is dense in the set of real numbers, i.e. between any two different real numbers there is a rational and an irrational number. *It follows that every real number is an accumulation point of both* $\mathbb{Q}$ *and* $\mathbb{R} \setminus \mathbb{Q}$*!*

• Definition (Dirichlet's function).

$$\mathcal{D} : \mathbb{R} \to \mathbb{R}, \qquad \mathcal{D}(x) := \begin{cases} 1, & \text{if } x \in \mathbb{Q}, \\ 0, & \text{if } x \in \mathbb{R} \setminus \mathbb{Q}. \end{cases}$$

• **Proposition.** *Dirichlet's function is discontinuous at any point of its domain.*

Proof. Let $p \in \mathbb{R}$. Then we can distinguish two complementary cases: either p is rational or irrational. Assume that p is rational (irrational case can be treated similarly). Then $\mathcal{D}(p) = 1$. On the other hand, the set of irrationals is dense in $\mathbb{R}$, hence there exists $(x_n) \subseteq \mathbb{R} \setminus \mathbb{Q}$ such that $x_n \to p$ if $n \to \infty$. Then, obviously, we have $\mathcal{D}(x_n) = 0$ for all $n \in \mathbb{N}$. Thus we obtain that

$$\lim_{n\to\infty} \mathcal{D}(x_n) = 0 \neq 1 = \mathcal{D}(p) = \mathcal{D}\big(\lim_{n\to\infty} x_n\big),$$

which means that $\mathcal{D}$ cannot be continuous at p.

Continuity and operations

• **Theorem (Continuity and operations).** Let $f, g : D \to \mathbb{R}$ be continuous functions at $p \in D$ and let $\lambda \in \mathbb{R}$ be any constant. Then the following statements hold.

- ⋆ The sum $f + g : D \to \mathbb{R}$ of f and g is continuous at p.
- ⋆ The product $f \cdot g : D \to \mathbb{R}$ of f and g is continuous at p. Consequently, λf is continuous at p.

• Notice that the converse of the statements is not true! Even if the functions $\mathcal{D} + (-\mathcal{D})$ and $0 \cdot \mathcal{D}$ are continuous, $\mathcal{D}$ is not!

• **Theorem (Continuity of the Composite Function).** Let $D, E \subseteq \mathbb{R}$, $f : D \to \mathbb{R}$, $g : E \to \mathbb{R}$, and assume that $f(D) \subseteq E$. Assume further that f is continuous at $p \in D$ and g is continuous at $f(p) \in E$. Then $g \circ f : D \to \mathbb{R}$ is continuous at p.

• **Theorem (Continuity of the Ratio Function).** Let $f, g : D \to \mathbb{R}$ be continuous at $p \in D$ and assume that $g(p) \neq 0$. Then $\frac{f}{g}$ is continuous at p.

Continuity on intervals

• **Theorem (Bolzano Theorem).** Continuous image of an interval is an interval.

• So continuous functions do not "tear apart" connected sets. If they take two different values on an interval, they take all the values between them!

• **Corollary.** If a function is continuous over an interval and takes both negative and positive values, then it has at least one zero in the interval.

Basic topological concepts

• **Definition (Open and closed sets).** The set $A \subseteq \mathbb{R}$ is called *open* if, for all $a \in A$, one can find $r > 0$ such that $]a - r, a + r[\subseteq A$. The set A is *closed* if $\mathbb{R} \setminus A$ is open.

• **Theorem (Characterization of global continuity).** A function $f : \mathbb{R} \to \mathbb{R}$ is continuous on $\mathbb{R}$ if and only if, for any open set $A \subseteq \mathbb{R}$,

$$f^{-1}(A) := \{x \in \mathbb{R} \mid \text{ létezik } y \in A, \text{ hogy } f(x) = y\}$$

is open. *("Preimage of any open set is open.")*

• **Definition (Compact sets).** The set $K \subseteq \mathbb{R}$ is called *compact* if it is closed and bounded.

• The interval $[0, 1]$ or any singleton $\{x\} \subseteq \mathbb{R}$ is compact.

• The interval $[0, 1[$ or $\mathbb{N}$ are not compact.

• In many respects, compact sets behave like finite sets and can be considered generalizations of them.

Continuity on compact sets

• **Theorem.** Continuous image of a compact set is compact.

• **Definition (Extreme values).** We say that $p \in D$ is a *place of minimum (place of maximum) of* $f : D \to \mathbb{R}$ *on* D if $f(p) \leq f(x)$ ($f(x) \leq f(p)$) holds for all $x \in D$. Then the value $f(p)$ is called the *minimum (maximum)* of f on D.

• **Proposition (Weierstrass Theorem).** On a non-empty compact set, a continuous function takes its minimum and maximum.

Proof. If $K \subseteq \mathbb{R}$ is compact and $f : K \to \mathbb{R}$ is continuous then $f(K)$ is compact, that is, it is closed and bounded. Because of its boundedness, it has an infimum and a supremum. And because it is closed, it also contains them.

Uniformly continuius functions

• **Definition (Uniform continuity).** We say that $f : D \to \mathbb{R}$ is uniformly continuous on D if, for all $\varepsilon > 0$, there exists $r > 0$ such that

$$|f(x) - f(y)| < \varepsilon, \qquad \text{whenever } |x - y| < r.$$

• Note that uniform continuity, unlike continuity, is not a local concept.

• Uniform continuity is stronger than continuity: any uniformly continuous function is continuous.

• There exists a continuous function that is not uniformly continuous.

Example. Let $f :]0, 1] \to \mathbb{R}$, $f(x) := \frac{1}{x}$. Obviously, f is continuous. Let $\varepsilon > 0$. Let us divide the interval $]0, 1]$ into n equal parts, where we do not yet say how large n is. The nodes are: $\{\frac{1}{n}, \frac{2}{n}, \dots, \frac{n-1}{n}, 1\}$. Then

$$\left|f\left(\frac{1}{n}\right) - f\left(\frac{2}{n}\right)\right| = \left|n - \frac{n}{2}\right| = \frac{n}{2}.$$

If f were uniformly continuous, then there would exist $r > 0$ such that $\frac{n}{2} < \varepsilon$, whenever $|\frac{1}{n} - \frac{2}{n}| = \frac{1}{n} < r$. This leads to a contradiction.

Uniformly continuous functions

- So in general, continuous functions are not uniformly continuous.
- **Theorem.** Continuous functions on compact sets are uniformly continuous.

Continuity of rational fractions

• **Definition (Rational fraction).** The function $f : D \to \mathbb{R}$ is called a rational fraction if there exist polynomials $p, q : \mathbb{R} \to \mathbb{R}$ such that $f(x) = \frac{p(x)}{q(x)}$ holds for all $x \in D$.

• **Continuity of rational fractions.** Keeping the notation of the above definition, a point $p \in D$ is a discontinuity of a rational fraction f if and only if $q(p) = 0$.

• So, to find the discontinuities of a rational fraction, we need to solve an algebraic equation. This can still be difficult if the polynomial in the equation is of degree five or higher. However, if the number of degrees is not greater than four, the problem can be solved.

• Later we will classify the discontinuities in general and see that a rational fraction can only have special discontinuities. Their type will be determined by the roots of the numerator (a problem of similar difficulty to the previous one).